

Solution Requirements Specification

1. Functional Requirements (FR)

Functional requirements detail the specific operations the ShopEEZ software system must execute:

FR ID	Feature Domain	Functional Spec Description	API Mapping
FR-01	Auth Control	Register new users checking password strength and unique emails.	/api/auth/register
FR-02	Auth Control	Store security password using Bcrypt salts inside database.	/api/auth/login
FR-03	Products Search	Search products by text, filter by category and range.	/api/products
FR-04	Persistent Cart	Serialize cart state in LocalStorage; load into Redux at bootstrap.	Client-side Redux store
FR-05	Checkout Pipeline	Calculate totals combining subtotal, 18% GST, and Flat Rs. 100 shipping.	/api/orders
FR-06	Product Reviews	Logged in customers leave reviews and star ratings on products.	/api/products/:id/reviews
FR-07	Admin Control	Dashboard aggregates metrics: orders count, products, sales revenue.	/api/admin/metrics
FR-08	Admin CRUD	Admin adds/updates products with multi-image Multer uploads.	/api/admin/products
FR-09	Stock Alerting	Mark products with stock <= 5 with low-stock warnings.	/api/admin/products

2. Non-Functional Requirements (NFR)

Non-functional criteria defining quality parameters, load performance, and database parameters:

NFR ID	Scope / Dimension	Target Performance Specification	Architectural Implementation
NFR-01	Security	All API routes (private profiles, checkouts, admin) must request JWT auth.	Express authorization middleware checking JWT header.
NFR-02	Data Integrity	Shopping cart state must persist across power cutoffs.	Redux persistence wrapper linking client LocalStorage.
NFR-03	Performance	Database calls for product searches must complete within 200ms.	Configured indexes in MongoDB Mongoose model on search keys.